Feedback Delay Network (FDN) Reverb Matrices

Unitary Matrix Feedback Math for `[reverb~]` Objects

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Abstract

This paper formulates the multi-channel Feedback Delay Network (FDN) reverberation architecture, Householder and Hadamard orthogonal feedback matrices, prime delay length distribution, and absorbing lowpass filter equations implemented in the `[reverb~]` node object.

1 Feedback Delay Network (FDN) Core Structure

An N-channel Feedback Delay Network consists of N parallel delay lines of prime sample lengths $\mathbf{m} = [m_1, m_2, \ldots, m_N]^T$, absorbing first-order lowpass filters $\mathbf{H}(z) = \mathrm{diag}(h_1(z), \ldots, h_N(z))$, and an $N \times N$ feedback matrix $\mathbf{A}$:

$$\mathbf{s}(n) = \mathbf{x}(n) + \mathbf{A} \cdot \mathbf{y}_{\text{delayed}}(n) \tag{1}$$

$$y_k(n) = h_k(z) * s_k(n - m_k), \quad k = 1, \ldots, N \tag{2}$$

2 Unitary Orthogonal Feedback Matrices A

To guarantee absolute energy conservation and prevent exponential energy growth or unnatural decay, the feedback matrix $\mathbf{A}$ must be orthogonal ($\mathbf{A}^T\mathbf{A} = \mathbf{I}$), ensuring eigenvalues lie on the unit circle ($|\lambda_i| = 1$).

2.1 Householder Reflection Matrix

The $N \times N$ Householder matrix is constructed as:

$$\mathbf{A}_{\text{Householder}} = \mathbf{I} - \frac{2}{N}\mathbf{1}_{N\times N} = \begin{bmatrix} 1-\frac{2}{N} & -\frac{2}{N} & \cdots & -\frac{2}{N} \\ -\frac{2}{N} & 1-\frac{2}{N} & \cdots & -\frac{2}{N} \\ \vdots & \vdots & \ddots & \vdots \\ -\frac{2}{N} & -\frac{2}{N} & \cdots & 1-\frac{2}{N} \end{bmatrix} \tag{3}$$

For $N = 4$ channels:

$$\mathbf{A}_4 = \begin{bmatrix} 0.5 & -0.5 & -0.5 & -0.5 \\ -0.5 & 0.5 & -0.5 & -0.5 \\ -0.5 & -0.5 & 0.5 & -0.5 \\ -0.5 & -0.5 & -0.5 & 0.5 \end{bmatrix} \tag{4}$$

3 Prime Delay Length Selection & Modal Density

Delay lengths m_k are chosen as mutually co-prime integers (e.g. $m \in \{1087, 1297, 1609, 1933\}$ at 44.1 kHz) to eliminate standing periodic comb echoes and maximize echo density growth rate:

$$\text{Density}(t) \propto \frac{t^{N-1}}{\prod_{k=1}^{N} m_k} \tag{5}$$

4 Frequency-Dependent Absorption & Reverb Time T_{60}

High frequencies decay faster in physical rooms due to air absorption. Each delay line includes a 1-pole lowpass filter $h_k(z) = \frac{1-g_k}{1-g_k z^{-1}}$ where gain g_k is computed from target reverberation time T_{60}:

$$g_k = 10^{-\frac{3m_k}{f_s \cdot T_{60}}} \tag{6}$$